

Experiment No: 13

Aim:

To design and Implement a digital clock using Verilog HDL in Xilinx Vivado.

Apparatus required:

↳ System

↳ Xilinx Vivado.

Procedure:

- 1) Open Xilinx Vivado.
- 2) Create a new RTL Project.
- 3) Add the Verilog Source file.
- 4) Add the testbench file.
- 5) Run Behavioural simulation.
- 6) Simulate and Verify the output waveform.

Programming Code:

Verilog Code

module digital_Clock(

input clk,

input rst,

output reg [5:0] sec,

);

```

always @(posedge clk (or) posedge rst)
begin
    if (rst)
        sec <= 0;
    else
        sec <= sec + 1;
end
end module.

```

Timescale - 1ns/1ps

```

module tb-digital-clock;
reg clk, rst;
wire cfb[5:0] sec;
digital_clock uut
    .clk(clk),
    .rst(rst),
    .sec(sec);
);

```

```

always #5 clk = ~clk;
initial design
    clk = 0;
    rst = 1;
    #10 rst = 0;
    #760
    $finish;
end

```

end module



Result.

Flow
and The digital clock was successfully designed
and simulated in xilinx Vivado

Experiment No. 14

Design and Simulate CMOS Inverter using Cadence design flow.

Aim: To design and simulate the CMOS inverter using Cadence virtuoso and analyse its DC transfer characteristics and transient response

Objectives:

- To design CMOS inverter schematic
- To perform DC sweep analysis
- To perform transient analysis
- To determine switching threshold & propagation delay

Software and tools used:

- Cadence virtuoso
- Spectra Simulator
- CMOS technology library

Procedure:

- 1) open Cadence Virtuoso
- 2) Create a new Library and attach (Mos file)
- 3) Create new Cell view (Schematic)
- 4) place PMOS, NMOS, transistor
- 5) Connect PMOS to VDD, NMOS to GND and Gates together as V_{in} , Drains together as V_{out}
- 6) Add DC Voltage Source
- 7) Check and Save
- 8) open ADE
- 9) select DC Sweep analysis
- 10) sweep V_{in} from 0V to VDD
- 11) Plot V_{out} Vs V_{in}
- 12) Replace DC input with pulse Source
- 13) Select transient analysis.
- 14) Run Simulation.
- 15) Plot V_{in} and V_{out} .

Calculations

Propagation delay:

$$t_p = (t_{pLH} + t_{pHL})/2$$



Result:

The CMOS Inverter was successfully designed and simulated using Cadence Virtuoso. The DC and transient analysis were performed and the switching characteristics were verified.

Experiment - 15

Design and DC of transient analysis of a CMOS all gate ~~NAND~~ gates using Cadence Virtuoso.

Aim:

To design and Simulate a 2-input CMOS all gate using Cadence Virtuoso and analyze in DC and transient characteristics.

Objectives:

- L> To design of 2input CMOS all gate Schematic
- L> To understand pullup and pull down network
- L> To perform DC Sweep analysis.

Software and tools used:

- L> Cadence Virtuoso
- L> Spectre Simulator
- L> CMOS technology Library

Procedure:

- 1) Open Cadence Virtuoso
- 2) Create new Library and attach CMOS technologyfile
- 3) Create a new schematic Cell view
- 4) place
 - L> 2 PMOS and 2 NMOS transistors
- 5) Add VDD and GND · synthesis.

- 6) Check and save
- 7) open ADE
- 8) Select DC sweep
- 9) Observe output voltage
- 10) Apply pulse input A to B
- 11) Select transient response / Analysis
- 12) Plot A, B, Y
- 13) Measure Propagation Delay

Calculations:

$$t_p = (t_{PLH} + t_{PHL}) / 2$$

Result:

The CMOS ~~NAND~~^{All} gate was successfully designed and simulated. the output matches the all logic truth table and dynamic characteristics were Verified